

First Order Logic (FOL)

domain / universe: the set where the elements come from

Predicate / function $P(x)$: the statement is true for variable x

$\forall$ for all

$\exists$ for some (there exists one)

$$U = \{x_1, x_2, \dots, x_n\}$$

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

$\hookrightarrow P(\cdot)$ holds for all elements of U

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

$\hookrightarrow P(\cdot)$ holds for some element of U

There is an element of U for which $P(\cdot)$ holds

exmp: some babies are cute

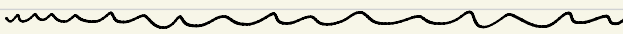
$U = \{\text{all babies}\}$

$C(x)$: x is cute

$\exists x C(x)$

all babies are cute

$\forall x C(x)$



an alternative solution

$U = \{\text{all people}\}$

$C(x)$: x is cute

$B(x)$: x is a baby

$\exists x (B(x) \wedge C(x))$

all babies are cute

$\forall x (B(x) \wedge C(x))$: everyone is a cute baby

$\forall x (B(x) \Rightarrow C(x))$

No baby is cute

$\sim \forall x (B(x) \Rightarrow C(x))$ X

$\forall x (B(x) \Rightarrow \sim C(x))$

some babies are not cute

De Morgan:

$$\sim \forall x P(x) \equiv \exists x \sim P(x)$$

$$\sim \exists x P(x) \equiv \forall x \sim P(x)$$

short proof

$$U = \{x_1, x_2, \dots, x_n\}$$

$$\sim \forall x P(x) \equiv \sim (P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n))$$

$$\equiv \sim P(x_1) \vee \sim P(x_2) \vee \dots \vee \sim P(x_n)$$

$$\equiv \exists x \sim P(x)$$

D.M. from

$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

$$\sim \forall x (B(x) \Rightarrow C(x)) \equiv \exists x \sim (B(x) \Rightarrow C(x)) \quad \text{D.M.}$$

$$\equiv \exists x \sim (\sim B(x) \vee C(x)) \quad \text{Implication}$$

$$\equiv \exists x (\sim \sim B(x) \wedge \sim C(x)) \quad \text{D.M.}$$

$$\equiv \exists x (B(x) \wedge \sim C(x))$$

Some babies are not cute

Inference Rules for FOL

$\forall x P(x)$

$P(c)$

UI (Universal Instantiation)

UE (Universal Elimination)

c : variable (refers to every element of universe)

c : constant (" " one element of universe)

$P(c)$

c is variable

Universal Generalization (UG)

$\forall x P(x)$

$\exists x P(x)$

$P(c)$

Existential Instantiation (EI)

Existential Elimination (EE)

c is a constant

$P(c)$

$\exists x P(x)$

Existential Generalization (EG)

c is constant or variable

Prove that $\underbrace{\exists x (d(x) \wedge \beta(x))}_{\text{Hyp.}} \Rightarrow \underbrace{\exists x d(x) \wedge \exists x \beta(x)}_{\text{Conc.}}$

1. $\exists x (d(x) \wedge \beta(x))$ Hyp.

2. $d(c) \wedge \beta(c)$ EI (1) c : constant

3. $d(c)$ Simp (2)

4. $\beta(c)$ Simp (2)

5. $\exists x d(x)$ EG (3)

6. $\exists x \beta(x)$ EG (4)

7. $\exists x d(x) \wedge \exists x \beta(x)$ Conj (5) + (6)

Prove that if $\underbrace{\exists x (d(x) \Rightarrow \beta(x)) \wedge \forall x d(x)}_{\text{Hyp.}}$ then $\underbrace{\exists x \beta(x)}_{\text{Conc.}}$.

1. $\exists x (d(x) \Rightarrow \beta(x)) \wedge \forall x d(x)$ Hyp.

2. $\exists x (d(x) \Rightarrow \beta(x))$ Simp (1)

3. $\forall x d(x)$ Simp (1)

4. $d(c) \Rightarrow \beta(c)$ EE (2) c : constant

5. $d(c)$ UI (3) (reusing constant c)

6. $\beta(c)$ MP (4) + (5)

7. $\exists x \beta(x)$ EG (6)

1. $\exists x \beta(x)$

2. $\exists x d(x)$

3. $\beta(c)$ EI① c : constant

~~4. $d(c)$ EI② WRONG~~

4. $d(k)$ EI② k : constant

Given: $\forall x (P(x) \Rightarrow Q(x))$

$$\exists x (R(x) \wedge \forall y (Q(y) \Rightarrow \sim S(x,y)))$$

Prove: $\exists x (R(x) \wedge \forall y (P(y) \Rightarrow \sim S(x,y)))$